

2.1 Sets

Monday, September 30, 2024 7:51 AM

Definition: A set is an unordered collection of distinct objects called elements.

We write $a \in A$ to denote "a" is an element of the set A. $a \notin A$ means a is not "in" A.

Definition: Two sets A and B are equal if they have the same elements. We write $A=B$. Then $A=B$ iff

$$\forall x (x \in A \wedge x \in B)$$

Ways To Express Sets

Roster	Set-Builder
List every element within braces $\{ \}$	Describe universal properties of an arbitrary element
$\{1, 2, 4, -2\}$	$\{x \mid x \text{ has property } P\}$
$\{a, b, c, d\}$	

The integers between 1 and 100 (inclusive) may be written $\{1, 2, \dots, 99, 100\}$ using a roster method, or as

$\{x \mid x \in \mathbb{Z} \wedge (1 \leq x \leq 100)\}$ using set-builder,

$\mathbb{N}$ = set of natural numbers.

$\mathbb{N} = \{0, 1, 2, 3, \dots\}$ book includes 0.

mathematicians tend not to. I will try matching the book.

$\mathbb{Z}$ = set of integers = $\{\dots, -2, -1, 0, 1, 2, \dots\}$

$\mathbb{Z}^+$ = set of positive integers

$\mathbb{Z}^+ = \{1, 2, 3, \dots\} = \{x \mid x \in \mathbb{Z} \wedge (x > 0)\}$

$\mathbb{Q}$ = set of rational numbers

$\mathbb{Q} = \{\frac{p}{q} \mid p, q \in \mathbb{Z} \wedge (q \neq 0)\}$

Since the definition says "distinct objects"

$$\{1, 1, 2, 3, 3, 3\} = \{1, 2, 3\}$$

$$\text{but } \{1, \{1\}, 2, 3\} \neq \{1, 2, 3\}$$

because the second 1, is distinct in that it is a set containing 1.

$$\begin{aligned}\mathbb{Q} &= \text{set of rational numbers} \\ \mathbb{Q} &= \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z} \wedge (q \neq 0) \right\} \\ \mathbb{Q}^+ &= \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z} \wedge (p > 0) \wedge (q > 0) \right\} \\ &= \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}^+ \right\}\end{aligned}$$

$$\begin{aligned}\mathbb{R} &= \text{set of real numbers} \\ \mathbb{R}^+ &= \{x \mid x \in \mathbb{R} \wedge (x > 0)\}\end{aligned}$$

$$\begin{aligned}\mathbb{C} &= \text{set of complex numbers} \\ \mathbb{C} &= \{a + bi \mid a, b \in \mathbb{R}\}\end{aligned}$$

Recall interval notation!

$$[a, b] = \{x \mid x \in \mathbb{R} \wedge (a \leq x \leq b)\}$$

$$(a, b) = \{x \mid x \in \mathbb{R} \wedge (a < x < b)\}$$

Definition: The empty set (or null set) is the set with no elements. The empty set is denoted by $\emptyset$ or $\{\}$.

Definition: A singleton set is a set with only one element.

Since $\emptyset$ is a set, $\{\emptyset\}$ is a non-empty singleton set containing the empty set.

Think of $\emptyset$ as an empty folder in a file directory. $\emptyset$ is just an empty folder but $A = \{\emptyset\}$ is a folder with a single folder inside which happens to be empty.

Definition: Let A and B be sets.

If every element of A is also an element of B ($\forall x (x \in A \Rightarrow x \in B)$) then we write $A \subseteq B$. Then A is a subset of B and B is a superset of A .

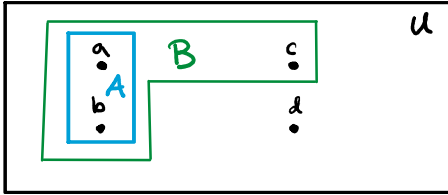
$$A \subseteq B \rightarrow \text{"A is contained in B"}$$

$$B \supseteq A \rightarrow \text{"B contains A"}$$

Venn Diagrams

We can visualize sets, subsets, and super sets using Venn diagrams.

Let $U = \{a, b, c, d\}$, $A = \{a, b\}$,
 $B = \{a, b, c\}$



So we can visualize $A \subseteq B \subseteq U$.

To show $A \subseteq B$, we show that for an arbitrary $x \in A$, $x \in B$.

To show $A \not\subseteq B$, it is enough to find an example of one $x \in A$ where $x \notin B$.

To show $A = B$, we can show $A \subseteq B$ and $B \subseteq A$.

Definition: The cardinality of a set A is the number of elements in A. We denote the cardinality of A by $|A|$.

Definition: A set is infinite if it contains infinitely many elements. ($|A| = \infty$)

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ are all infinite sets.

If $A = \{a, b, c, d, \dots, x, y, z\}$, then
 $|A| = 26$.

Definition: For a set A, the power set of A is the set of all subsets of A.

Denoted $\mathcal{P}(A)$. $\rightarrow$ Script P.

If $|A| = n$, $|\mathcal{P}(A)| = 2^n$.